

- **Chorionic Villus Sampling (CVS)** – placental tissue sample
- **Fetal Blood Sampling (Cordocentesis)** – blood from umbilical cord (*rare and high-risk*)

Comparison of Fetal Screening Technologies

Method	Type	When Done	Sample	Benefits	Limitations / Risks
Carrier Testing	Non-Invasive	Before conception	Blood or saliva	No risk, identifies risk in parents	Doesn't test fetus directly
PGD (IVF)	Minimally Invasive	Before implantation	Embryo cell	Prevents affected embryo implantation	IVF needed, ethical concerns
Ultrasound	Non-Invasive	Throughout pregnancy	Imaging	Detects physical defects, safe	Can't detect genetic mutations
NIPT	Non-Invasive	~10 weeks onward	Maternal blood	No miscarriage risk, early detection	Screening only, false positives, confirm with invasive test
Amniocentesis	Invasive	15–20 weeks	Amniotic fluid	Accurate genetic diagnosis	1% miscarriage risk, discomfort
CVS	Invasive	10–13 weeks	Placental tissue	Earlier results than amniocentesis	1–2% miscarriage risk
Fetal Blood Sampling	Invasive	~18+ weeks (rare)	Fetal blood (cord)	Confirms disorders when other tests unclear	Highest miscarriage risk (~2–4%)

Note: There is a chance for all methods to be **false positive** or **false negative**.

Edexcel IAL Biology Unit 1: Key Definitions Table

Term	Definition
Monosaccharide	A single sugar unit; the simplest type of carbohydrate (e.g., glucose, fructose).
Disaccharide	A carbohydrate formed by two monosaccharides joined by a glycosidic bond (e.g., sucrose, maltose).
Polysaccharide	A complex carbohydrate formed by many monosaccharides linked by glycosidic bonds (e.g., starch, cellulose).
Glycosidic bond	A covalent bond that joins carbohydrate (sugar) molecules together, formed by a condensation reaction.

Condensation reaction	A chemical reaction in which two molecules combine to form a larger molecule, producing water as a byproduct.
Hydrolysis	A reaction that breaks down molecules by adding water, breaking covalent bonds (e.g., glycosidic or peptide bonds).
Triglyceride	A lipid molecule composed of one glycerol and three fatty acids, used mainly for energy storage.
Saturated fatty acid	A fatty acid with no carbon-carbon double bonds; typically solid at room temperature.
Unsaturated fatty acid	A fatty acid with one or more carbon-carbon double bonds; usually liquid at room temperature.
Atherosclerosis	A disease in which plaque builds up inside arteries, narrowing them and reducing blood flow.
Thrombosis	The formation of a blood clot (thrombus) within a blood vessel, obstructing blood flow.
Coronary heart disease (CHD)	A type of cardiovascular disease where the coronary arteries become narrowed due to atherosclerosis, reducing oxygen supply to the heart muscle.
Cardiovascular disease (CVD)	A general term for conditions affecting the heart or blood vessels, usually associated with atherosclerosis and thrombosis.
Blood pressure	The force of blood against the walls of arteries; measured as systolic over diastolic pressure.
Systolic pressure	The pressure in the arteries during contraction of the heart muscle.
Diastolic pressure	The pressure in the arteries when the heart muscle is relaxed between beats.
Enzyme	A biological catalyst, usually a protein, that speeds up the rate of a specific biochemical reaction.
Active site	The region on an enzyme where the substrate binds and the reaction is catalyzed.
Substrate	The specific reactant that an enzyme acts upon.
Lock and key model	Model where the substrate fits exactly into the enzyme's active site like a key into a lock.
Induced fit model	Model where enzyme changes shape slightly to accommodate the substrate, improving binding.
Diffusion	The net movement of particles from an area of high concentration to an area of low concentration.

Osmosis	The net movement of water molecules from a region of high water potential to low water potential through a partially permeable membrane.
Active transport	The movement of substances against a concentration gradient, requiring energy from ATP.
Endocytosis	The process of taking substances into the cell by engulfing them in a vesicle formed from the plasma membrane.
Exocytosis	The process by which substances are released from a cell through vesicles fusing with the plasma membrane.
Plasma membrane	A partially permeable membrane surrounding the cytoplasm, controlling movement of substances into and out of the cell.
Phospholipid bilayer	A double layer of phospholipids that forms the basic structure of cell membranes.
Cholesterol	A lipid molecule within the cell membrane that stabilizes its fluidity and mechanical strength.
Glycoprotein	A protein with carbohydrate groups attached; involved in cell recognition and signaling.
Glycolipid	A lipid with carbohydrate attached; involved in cell recognition.
Carrier protein	A membrane protein that changes shape to transport substances across membranes.
Channel protein	A membrane protein that provides a passageway for specific substances to diffuse through the membrane.
Antioxidant	A molecule that inhibits oxidation and neutralizes free radicals, preventing cellular damage.
Risk factor	Any attribute, characteristic, or exposure that increases the likelihood of developing a disease or injury.
Correlation	A statistical relationship or association between two variables.
Causation	A relationship where one event is the result of the occurrence of another event (i.e., cause and effect).
Clinical trial	A controlled experiment carried out on humans to test the effectiveness and safety of drugs or treatments.
Placebo	A substance with no therapeutic effect, used as a control in testing new drugs.
Double-blind trial	A clinical trial in which neither the participants nor the researchers know who is receiving the active treatment or the placebo.

Randomized trial	A clinical study where participants are randomly assigned to different treatment groups.
Validity	The extent to which an experiment measures what it claims to measure.
Reliability	The consistency and repeatability of experimental results.
Bias	A systematic error introduced into sampling or testing that affects results.